Structured questions

|!|BCELB0230400001|!| ★ **Cross topic** **STSE**

The Food and Environmental Hygiene Department (FEHD) monitors the populations of the mosquito *Aedes albopictus* in various areas in Hong Kong and releases the monthly ovitrap index (MOI). The table below shows the monthly ovitrap indices in Tseung Kwan O (North) in 2018.

	Jan	**Feb**	**Mar**	**Apr**	**May**	**Jun**
MOI	0.0%	0.0%	1.6%	9.8%	32.8%	41.0%

	Jul	**Aug**	**Sept**	**Oct**	**Nov**	**Dec**
MOI	33.3%	16.8%	3.4%	5.1%	0.0%	0.0%

(a) Mosquitoes are the vectors for many diseases such as malaria. Name another disease that is transmitted by mosquitoes. (1 mark)

(b) Explain why the human body's physical barriers do not prevent the entry of pathogens transmitted by mosquitoes. (2 marks)

(c) With reference to the information provided in the table, describe how ***one*** factor affects the breeding of mosquitoes. (2 marks)

(d) The FEHD advises actions to be taken when the ovitrap index is higher than 20%. Describe ***two*** mosquito control measures that you can take to

(i) prevent mosquito bites. (2 marks)

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(ii) prevent mosquito proliferation. (2 marks)

##

(a) Dengue fever / Japanese encephalitis / Zika fever (1)

(b) Mosquitoes bite on the human skin to feed on blood. (1)

This pierces a hole on the skin through which pathogens may enter. (1)

(c) The ovitrap index is higher in the summer (from May to August) and is lower during winter (from November to February). (1)

A higher temperature favours the breeding of mosquitoes. (1)

(d) (i) Wear loose, light-coloured, long-sleeved tops and trousers. (1)

Use (DEET-containing) insect repellent on exposed parts of the skin and clothing. (1)

(ii) Kill mosquitoes by spraying pesticides or larvicidal oil around mosquito's habitats. (1)

Remove the potential breeding grounds of mosquitoes, e.g. clear standing water which is a common breeding site for mosquitoes / change the water in vases once a week / avoid using saucers underneath flower pots / cover water containers tightly / ensure air-conditioner drip trays are free of stagnant water / put all used cans and bottles into covered dustbins. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	2
(c)	Analysing data	2
(d)(i)	Understanding basic concepts	2
(d)(ii)	Understanding basic concepts	2

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|!|BCELB0230500002|!| ★ **STSE**

Avian influenza is caused by the types of influenza viruses that mainly affect birds, such as chickens and ducks. Human cases infected with avian influenza A viruses (e.g. H5N1, H5N6, H7N9, etc.) have been identified in recent years.

(a) Human-to-human transmission of avian influenza is inefficient. Suggest ***two*** routes by which avian influenza might be transmitted from poultry to humans. (4 marks)

(b) Based on the transmission routes you have given in (a), suggest ***two*** preventive measures for avian influenza. (2 marks)

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(c) Explain whether antibiotics can be used to treat human cases of avian influenza. (2 marks)

##

(a) Infected poultry and their droppings contain avian influenza viruses. (1)
If a person has touched infected poultry or their droppings, the viruses may enter his/her body through wounds on the skin (or when he/she rubs the eyes and nose with the contaminated hands). (1)
The eggs and meat of poultry may be contaminated with avian influenza viruses. (1)
If a person has eaten the contaminated raw eggs or meat, the viruses may enter his/her body through the alimentary canal. (1)
(Or other reasonable answers)

(b) Avoid touching poultry in wet markets and wild birds. (1)
Cook poultry eggs and meat thoroughly. (1)
(Or other relevant answers)

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(c) No, antibiotics are ineffective against human cases of avian influenza (1)
because the disease is caused by viruses. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	4
(b)	Applying concepts	2
(c)	Applying concepts	2

|!|BCELA0230400003|!| ★★ **STSE**

Middle East respiratory syndrome (MERS) is a viral respiratory disease caused by a coronavirus (MERS-CoV) that was first identified in Saudi Arabia in 2012. Coronaviruses are a large family of viruses which include viruses that cause mild illness such as the common cold as well as severe illness such as severe acute respiratory syndrome (SARS) in humans.

(a) Describe the typical structure of viruses. (2 marks)

The table below lists the number of MERS confirmed cases and deaths between June 2012 and January 2020.

	Confirmed cases	Deaths
Saudi Arabia	1029	452
South Korea	184	38
United Arab Emirates	74	10
Jordan	19	6
Qatar	10	4
Oman	5	3
Iran	5	2
United Kingdom	4	3
Germany	3	1
Kuwait	3	1
Tunisia	3	1
Algeria	2	1
France	2	1
Spain	2	0
Netherlands	2	0
Philippines	2	0
United States	2	0

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Greece	1	1
Malaysia	1	1
Turkey	1	1
Yemen	1	1
Austria	1	0
Egypt	1	0
Italy	1	0
Lebanon	1	0
Thailand	1	0

(b) Calculate the death rate in Saudi Arabia. Show your working. (2 marks)

(c) In recent years, novel diseases such as MERS and SARS could spread rapidly around the world. Suggest a factor that helps spread MERS to many countries rapidly. (1 mark)

Scientists have not yet fully understood the modes of transmission of MERS. People may be infected upon exposure to animals (especially dromedary camels) or close contact with other confirmed patients (such as healthcare workers providing unprotected care to patients).

(d) Based on the current understanding about the transmission of MERS, suggest ***two*** ways to prevent MERS. (2 marks)

##

(a) A virus is a non-cellular entity (1)
that consists of a protein coat surrounding genetic material (DNA or RNA). (1)

(b) 452 / 1029 × 100% (1)
= 44% (1)

(c) People may travel around the world frequently nowadays (for leisure or business trips) (1)

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(d) Any ***two*** of the following: (2)

- Anyone visiting farms / markets / other places where dromedary camels are present should wash their hands with soap and water before and after touching animals, and should avoid contact with sick animals.
- Avoid going to farms, barns and markets with camels.
- Camel meat should be properly cooked / Camel milk should be properly pasteurized before consumption.
- Anyone coming into contact with MERS patients should wear a medical mask, eye protection and protective clothing.
- Anyone coming into contact with MERS patients should wash their hands thoroughly with soap and water after contact with the patients.

(Or other reasonable answers)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Mathematical skills	2
(c)	Applying concepts	1
(d)	Applying concepts	2

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|!|BCELA0230400004|!| ★★ **STSE**

Read the following news report about food-borne infection due to *Salmonella* and answer the questions that follow.

200 cases of food-borne infection linked to sandwiches

May 30, 2020

The Centre for Health Protection (CHP) of the Department of Health reported that 229 people were sick and 45 needed hospital treatment after eating sandwiches which were produced by an unlicensed food factory in Hong Kong.

The CHP has collected food samples from the factory and stool samples from the patients for testing. Among all the samples, stool samples from 12 sick people and one food sample tested positive for *Salmonella*.

Salmonella are bacteria commonly found in eggs, dairy products and meat. After reaching the human intestines, *Salmonella* multiplies and produces toxins which damage and kill host cells, resulting in symptoms such as vomiting, diarrhoea and fever.

(a) State the transmission route of *Salmonella* in the cases described in the news report. (1 mark)

(b) The CHP reminds food manufacturers to make efforts to ensure food safety by following the guidelines listed below. Explain how each guideline can minimize the risk of food-borne infection.

(i) Cook food thoroughly. (1 mark)

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(ii) Store cooked food at the right temperature (e.g. at 4 °C). (2 marks)

(iii) Separate raw food and cooked food during handling and storage. (2 marks)

(c) Suggest ***two*** precautions that the public can take to reduce the risk of getting food-borne infection. (2 marks)

##

(a) Through the ingestion of food/sandwiches contaminated by *Salmonella* (1)

(b) (i) By cooking food thoroughly can reduce the chance of any living pathogens remaining on the food. (1)

(ii) At room temperature, bacteria multiply quickly. (1)
At lower temperatures such as 4 °C, the growth of bacteria can be greatly slowed down. (1)

(iii) Raw food may contain bacteria and other pathogens which may contaminate the cooked food. (1)
If the contaminated cooked food is eaten without heating up, the pathogens may enter the human body with the food. (1)

(c) Any ***two*** of the following: (2)

- Buy food from reliable sources.
- Consume cooked food / ready-to-eat food as soon as possible.
- Wash hands thoroughly with soap and water before handling food and eating.
- Store any leftover food at the right temperature (e.g. 4 °C) and heat up the food thoroughly before consumption.

(Or other reasonable answers)

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(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)(i)	Applying concepts	1
(b)(ii)	Applying concepts	2
(b)(iii)	Applying concepts	2
(c)	Applying concepts	2

|!|BCELA0230400005|!| ★★ **STSE**

Zika virus infection is a mosquito-borne disease caused by Zika virus. Zika virus is mainly transmitted to humans through the bites of the mosquito *Aedes aegypti*, which is currently not found in Hong Kong. Other *Aedes* mosquito species such as *Aedes albopictus*, which is commonly found in Hong Kong, are also considered as potential vectors. Zika virus has also been found in human semen and transmission by sexual contact has been confirmed.

(a) Describe ***two*** ways to prevent Zika virus infection by reducing mosquito proliferation. (2 marks)

(b) Suggest ***two*** ways to prevent sexual transmission of Zika virus infection. (2 marks)

(c) The World Health Organization has concluded that Zika virus infection during pregnancy is a cause of congenital brain abnormalities, including microcephaly (小頭症). Give advice to pregnant women on preventive measures against Zika virus infection. (2 marks)

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(d) Give another example of mosquito-borne disease. (1 mark)

##

(a) Any ***two*** of the following: (2)

- Spraying pesticides or larvicidal oil around mosquito's habitats
- Removing the potential breeding grounds of mosquitoes, such as clearing standing water / changing the water in vases once a week
- Covering water containers tightly to prevent mosquito breeding in the water

(Or other reasonable answers)

(b) Any ***two*** of the following: (2)

- Travellers should consider not having sex during travel to areas with Zika virus transmission, or else condom should be used.
- Travellers returning from areas with Zika virus transmission should abstain from sex for 3 months, or else condom should be used.

(Or other reasonable answers)

(c) Any ***two*** of the following: (2)

- Apply (DEET-containing) insect repellents to exposed areas of the skin to avoid mosquito bites
- Use mosquito nets or screens to avoid mosquito bites
- Avoid travelling to areas with Zika virus transmission
- Avoid having sex with her partner who had travelled to areas with Zika virus transmission, or else condom should be used throughout the pregnancy

(Or other reasonable answers)

(d) Malaria / Dengue fever / Japanese encephalitis / other correct answer (1)

(7 marks)

##

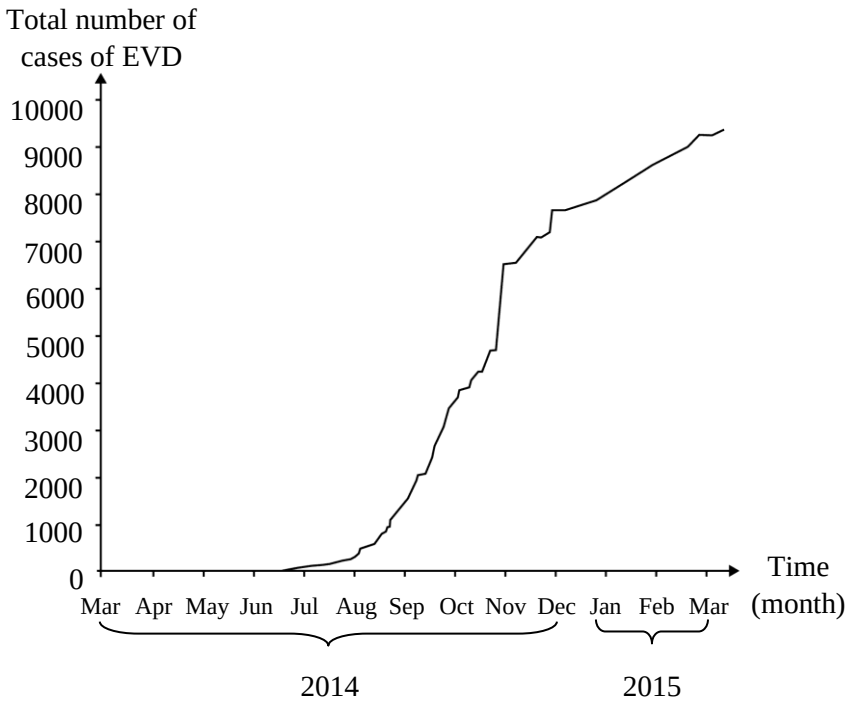
	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	2
(c)	Applying concepts	2
(d)	Understanding basic concepts	1

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|!|BCELB0230400006|!| ★ **STSE**

Ebola virus disease (EVD) is a severe disease in humans and it is often fatal. The virus causing this disease can be transmitted from wild animals to humans and can spread among humans through direct contact and body fluids.

The graph below shows the total number of cases of EVD in Liberia, a West African country, during a recent outbreak of the disease.



Data source: https://www.cdc.gov/vhf/ebola/outbreaks/2014-west-africa/cumulative-cases-graphs.html

(a) How do viruses cause diseases in humans? (2 marks)

_____

_____

_____

(b) Identify from the graph above the period during which the number of EVD cases increased the most rapidly. (1 mark)

_____

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(c) Explain how each of the following practices in the villages with EVD outbreaks had contributed to the rapid rise in the number of EVD cases.

(i) At the start of the outbreak, there was a severe lack of trained health workers in the affected villages and much of the nursing was carried out in the family homes or at local clinics. (2 marks)

(ii) As the outbreak spread, some residents left their villages to flee from the disease. (1 mark)

(iii) Local mourning and burial practices involved gathering at the family house to pay respects to the deceased. Prior to burial the deceased was usually bathed by close family members. (1 mark)

(d) Suggest ***two*** ways to control the spread of EVD. (2 marks)

##

(a) After a virus gets into a living cell (the host cell), the virus takes over the cell's metabolism to produce new virus particles. This may interfere with the host cell's normal cell function. (1)
The new formed viruses may damage the cell membrane of the host cell in order to leave the cell. (1)

(b) From around the end of October 2014 to the beginning of November 2014 (1)

(c) (i) The patients could not receive effective treatments and EVD could easily spread to other patients in the same house. (1)
No trained health workers were there to teach the villagers how to prevent the spread of EVD. (1)

(ii) The residents who left their villages might have already carrying the ebola viruses which spread to farther places as the residents fleed. (1)

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(iii) Mourning and burial practices brought people into close contact, and the viruse could spread from symptomless carriers. / Bathing the deceased may spread the viruses through contact with body fluids. (1)

(d) Any ***two*** of the following: (2)

- Avoid close contact with infected people
- Clean hands frequently with liquid soap or disinfectants
- Wear gloves when handling wounds and body fluids
- Avoid sharing objects that may be contaminated with blood
- Practise safe sex and wear a condom during sexual contact

(Or other reasonable answers)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	1
(c)(i)	Applying concepts	2
(c)(ii)	Applying concepts	1
(c)(iii)	Applying concepts	1
(d)	Applying concepts	2

|!|BCELA0230500007|!| ★★ **STSE** **Cross topic**

✂

Infections caused by methicillin-resistant *Staphylococcus aureus* (MRSA) are difficult to treat because these bacteria are resistant to beta-lactam antibiotics such as penicillin, amoxicillin and methicillin. Some strains of *S. aureus* produce beta-lactamase, an enzyme that breaks the structures of beta-lactam antibiotics.

(a) Beta-lactam antibiotics are bactericidal and act by inhibiting the synthesis of the peptidoglycan layer of bacterial cell walls. Explain how these antibiotics kill non-resistant bacteria. (3 marks)

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(b) Describe how the indiscriminate use of antibiotics may have speeded up the development of antibiotic resistance in a population of *Staphylococcus aureus*. (4 marks)

(c) Using your knowledge about enzymes, suggest how antibiotic treatments for MRSA can be made effective again. (2 marks)

(d) In hospitals, immuno-deficient patients with wounds are at a greater risk of the infection of MRSA. The number of patients infected with MRSA has increased in some hospitals. Suggest ***two*** ways to reduce the transmission of MRSA in hospitals. (2 marks)

##

(a) The bacterial cell walls become fragile due to the lack of peptidoglycan. (1)
The bacterial cell walls are not tough enough to limit the influx of water by osmosis into the bacterial cells. (1)
The bacterial cells eventually burst due to the excessive influx of water. (1)

(b) Within a population of *Staphylococcus aureus*, some bacteria have antibiotic resistance while others do not. (1)
The antibiotic resistant bacteria produce the enzyme beta-lactamase which breaks the structure of antibiotics and thus these bacteria are not killed, while the non-resistant bacteria are killed by the antibiotics. (1)
Facing less competition for resources, the antibiotic-resistant bacteria reproduce rapidly and their number increases. (1)
Finally, the whole population of bacteria becomes resistant to the antibiotic. (1)

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(c) Prescribe antibiotics together with a drug that inhibits the action of the enzyme beta-lactamase. (1)

The activity of beta-lactamase would be greatly reduced, thus the antibiotics would remain intact to exert bactericidal effect. (1)

(d) Any ***two*** of the following: (2)

- Manage wounds properly by keeping wounds sterilized and covering them with waterproof adhesive dressings.
- Wash hands with liquid soap and water before and after touching wounds.
- Perform hand hygiene frequently, especially after touching public installations such as handrails or door knobs.

(Or other reasonable answers)

(11 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	4
(c)	Applying concepts	2
(d)	Applying concepts	2

|!|BCELB0230500008|!| ★

✂

Some antibiotics (e.g. penicillin) inhibit the synthesis of cell walls in bacteria.

(a) (i) Explain how penicillin leads to the death of bacteria as a result of osmosis. (4 marks)

(ii) Explain why penicillin is not harmful to human cells. (1 mark)

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(b) Describe ***two*** other ways by which an antibiotic affects the growth of bacteria. (4 marks)

(c) *Staphylococcus aureus* is a bacterium that can cause skin infection, respiratory infection and food poisoning. Antibiotic resistance in *S. aureus* was uncommon when penicillin was first introduced in 1943. The original Petri dish on which Alexander Fleming observed the antibacterial activity of penicillin was growing a culture of *S. aureus*. By 1950, 40% of hospital *S. aureus* were penicillin-resistant; by 1960, this had risen to 80%. Explain how *S. aureus* may have evolved resistance to penicillin. (4 marks)

##

(a) (i) Penicillin prevents bacterial cell walls from forming properly during cell division, thus the cell walls become weakened. (1)
When the water potential of the cytoplasm of a bacterial cell is lower than that of the surroundings, (1)
water enters the bacterial cell by osmosis and the cell swells. (1)
As the cell walls influenced by penicillin are not strong enough to limit the influx of water, the excessive water eventually causes the cell to burst. (1)

(ii) Human cells have no cell walls. (1)

(b) Any ***two sets*** of the following: (4)

- Some antibiotics inhibit protein synthesis by binding to bacterial ribosomes. (1)
  The bacteria cannot synthesize proteins which maintain normal metabolism. (1)
- Some antibiotics inhibit the replication of nucleic acids / RNA / DNA in bacteria. (1)
  The bacteria cannot carry out cell division / multiply. (1)
- Some antibiotics destroy bacterial cell membranes. (1)
  The cell contents leak out and the cell dies. (1)

(Or other reasonable answers)

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(c) There are genetic variations in *S. aureus* population: some bacteria are resistant to penicillin while some are susceptible. (1)
When patients do not complete the whole course of penicillin prescription, (1)
the resistant bacteria survive and the susceptible ones are killed by penicillin. (1)
Those resistant bacteria reproduce rapidly and become prevalent in the population. (1)

(13 marks)

##

	Skills assessed	Marks
(a)(i)	Applying concepts	4
(a)(ii)	Applying concepts	1
(b)	Understanding basic concepts	4
(c)	Applying concepts	4

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